

DIFFERENTIATION

WORKEDOUT EXAMPLES

W.E - 1 : If $f(x) = \begin{vmatrix} x^2 & 2x+1 & 2x^3-3 \\ \sin x & \cos x & -1 \\ 2 & 3 & x \end{vmatrix}$ then

the coefficient of x in f(x)

Ans. -13

Sol: coefficient of x in f(x) is $f'(0)$

$$\therefore f'(x) = \begin{vmatrix} 2x & 2 & 6x^2 \\ \sin x & \cos x & -1 \\ 2 & 3 & x \end{vmatrix} + \begin{vmatrix} x^2 & 2x+1 & 2x^3-3 \\ \cos x & -\sin x & 0 \\ 2 & 3 & x \end{vmatrix}$$

$$+ \begin{vmatrix} x^2 & 2x+1 & 2x^3-3 \\ \sin x & \cos x & -1 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\Rightarrow f'(0) = -13$$

W.E - 2 : Find the differential coefficient of

$$\tan^{-1} \left(\frac{\sqrt{2ax - x^2}}{a - x} \right) \text{ w.r.t. } x$$

Sol: Put $x = 2a \sin^2 \theta$

$$\therefore y = \tan^{-1} \left(\frac{2a \sin \theta \cos \theta}{a \cos 2\theta} \right) = 2\theta = 2 \sin^{-1} \sqrt{\frac{x}{2a}}$$

$$\therefore \frac{dy}{dx} = 2 \cdot \frac{1}{\sqrt{1 - \frac{x}{2a}}} \times \frac{1}{\sqrt{2a}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{\sqrt{2ax - x^2}}$$

W.E - 3 : Sum of the series

$$1 + 2x + 3x^2 + \dots + nx^{n-1} \quad (x \neq 1) \quad (n \in N)$$

Sol: Consider $1 + x + x^2 + \dots + x^n = \frac{x^{n+1} - 1}{x - 1}$

d.w.r.t. 'x' on both sides

$$1 + 2x + 3x^2 + \dots + nx^{n-1} = \frac{(x-1)(n+1)x^n - (x^{n+1} - 1)}{(x-1)^2}$$

$$= \frac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2}$$

W.E - 4 :

Sum of the series $\sin \theta + 3 \sin 3\theta$

$$+ 5 \sin 5\theta + \dots + (2n-1) \sin (n-1)\theta$$

Sol: Consider

$$\cos \theta + \cos 3\theta + \cos 5\theta + \dots + \cos (2n-1)\theta$$

$$= \frac{\sin n\theta}{\sin \theta} \cos n\theta$$

$$= \frac{\sin 2n\theta}{2 \sin \theta}$$

differentiate w.r.t. θ on both sides

$$-\sin \theta - 3 \sin 3\theta - 5 \sin 5\theta - \dots - (2n-1) \sin (2n-1)\theta$$

$$= \frac{1}{2} \frac{\sin \theta \times 2n \cos 2n\theta - \sin 2n\theta \cos \theta}{\sin^2 \theta}$$

$$\Rightarrow \sin \theta + 3 \sin 3\theta + 5 \sin 5\theta + \dots + (2n-1) \sin (2n-1)\theta =$$

$$\frac{-2n \sin \theta \cos 2\theta + \sin 2n\theta \cos \theta}{2 \sin^2 \theta}$$

$$= \frac{-2n (\sin (2n+1)\theta - \sin (2n-1)\theta) + \sin (2n+1)\theta + \sin (2n)\theta}{(2 \sin \theta)^2}$$

$$= \frac{(2n+1)\sin(2n-1)\theta - (2n-1)\sin(2n+1)\theta}{2(1-\cos\theta)}$$

W.E - 5 : By using the sum of the series

$\sin\theta + \sin 2\theta + \sin 3\theta + \dots + \sin n\theta$ **prove**

that $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

Sol: $\sin\theta + \sin 2\theta + \sin 3\theta + \dots + \sin n\theta$

$$= \frac{\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)} \sin\left(\frac{(n+1)\theta}{2}\right)$$

$$= \frac{\cos\left(\frac{\theta}{2}\right) - \cos\left(\frac{(2n+1)\theta}{2}\right)}{2\sin\frac{\theta}{2}}$$

differentiate w.r.t. θ on both sides

$$\begin{aligned} & \cos\theta + 2\cos 2\theta + 3\cos 3\theta + \dots + n\cos n\theta \\ &= \frac{1}{2} \left[\frac{\sin\frac{\theta}{2} \left(-\sin\frac{\theta}{2} \right) \left(\frac{1}{2} \right) + \sin\left(\frac{(2n+1)\theta}{2}\right) \left(\frac{2n+1}{2} \right)}{\sin^2\frac{\theta}{2}} \right] \\ & \quad - \left(\cos\left(\frac{\theta}{2}\right) - \cos\frac{(2n+1)\theta}{2} \right) \cos\frac{\theta}{2} - \frac{1}{2} \\ &= \frac{1}{4} \left[\frac{-\sin^2\frac{\theta}{2} + (2n+1)\sin\frac{\theta}{2}\sin\frac{(2n+1)\theta}{2} - \cos^2\left(\frac{\theta}{2}\right) + \cos\frac{\theta}{2}\cos\frac{(2n+1)\theta}{2}}{\sin^2\frac{\theta}{2}} \right] \\ &= \frac{1}{4} \left[\frac{-1 + 2n\sin\frac{\theta}{2}\sin\frac{(2n+1)\theta}{2} + \cos n\theta}{\sin^2\frac{\theta}{2}} \right] \end{aligned}$$

$$= \frac{\frac{1}{4} \left[-1 + n(\cos n\theta - \cos(n+1)\theta) + \cos n\theta \right]}{\sin^2\frac{\theta}{2}}$$

$$= \left[\frac{(n+1)\cos n\theta - n\cos(n+1)\theta - 1}{2(1-\cos\theta)} \right]$$

$$\lim_{\theta \rightarrow 0} (\cos\theta + 2\cos 2\theta + 3\cos 3\theta + \dots + n\cos n\theta)$$

$$= \lim_{\theta \rightarrow 0} \left[\frac{(n+1)\cos n\theta - n\cos(n+1)\theta - 1}{2(1-\cos\theta)} \right]$$

$$= \frac{1}{2} n(n+1) \lim_{\theta \rightarrow 0} \frac{\sin(n+1)\theta - \sin n\theta}{\sin\theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{n(n+1)}{2} \frac{2\cos(2n+1)\frac{\theta}{2} \sin\left(\frac{\theta}{2}\right)}{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)} = \frac{n(n+1)}{2}$$

W.E - 6 : How will the equation

$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$ **be transformed**

if we introduce a new independent variable 't' by substituting $\cos t = x$

Sol: Put

$$\cos t = x \Rightarrow -\sin t \frac{dt}{dx} = 1 \Rightarrow \frac{dt}{dx} = -\frac{1}{\sin t}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{-1}{\sin t} \frac{dy}{dt}$$

$$\Rightarrow x \frac{dy}{dx} = \frac{-\cos t}{\sin t} \frac{dy}{dt} = -\cot t \frac{dy}{dt}$$

$$\therefore -x \frac{dy}{dx} = \cot t \frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(-\frac{1}{\sin t} \frac{dy}{dt} \right) = \frac{d}{dt} \left(-\frac{1}{\sin t} \frac{dy}{dt} \right) \left(-\frac{1}{\sin t} \right)$$

$$\therefore (1-x^2) \frac{d^2y}{dx^2} = \sin^2 t \left(\frac{-\cos t}{\sin^3 t} \frac{dy}{dt} + \frac{1}{\sin^2 t} \frac{d^2y}{dt^2} \right)$$

$$\therefore (1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y$$

$$= -\cot t \frac{dy}{dt} + \frac{d^2y}{dt^2} + \cot t \frac{dy}{dt} + y = 0$$

$$\Rightarrow \frac{d^2y}{dt^2} + y = 0$$

LEVEL - V

1. Suppose for a differentiable function f ,

$$f(0) = 0, f(1) = 1 \text{ and } f'(0) = 4 = f'(1).$$

If $g(x) = f(e^x) \cdot e^{f(x)}$ then $g'(0)$ is.

- A) 4 B) 8 C) 2 D) -2

2. If $y = \frac{(a-x)\sqrt{a-x} - (b-x)\sqrt{x-b}}{\sqrt{a-x} + \sqrt{x-b}}$, then

$\frac{dy}{dx}$ (wherever it is defined) is

- A) $\frac{x+(a+b)}{\sqrt{(a-x)(x-b)}}$ B) $\frac{2x-a-b}{2\sqrt{a-x}\sqrt{x-b}}$
C) $-\frac{(a+b)}{2\sqrt{(a-x)(x-b)}}$ D) $\frac{2x+(a+b)}{2\sqrt{(a-x)(x-b)}}$

3. If $(\sin x)(\cos y) = 1/2$, then $\frac{d^2y}{dx^2}$ at

$$\left(\frac{\pi}{4}, \frac{\pi}{4} \right) \text{ is}$$

- A) -4 B) -2 C) -6 D) 0

4. Let $g(x)$ be the inverse of an invertible function $f(x)$, which is differentiable for all real x , then $g''(f(x))$ equals

A) $-\frac{f''(x)}{(f'(x))^3}$

B) $\frac{f'(x)f''(x) - (f'(x))^3}{f'(x)}$

C) $\frac{f'(x)f''(x) - (f'(x))^2}{(f'(x))^2}$

D) zero

5. If $f(x)$ satisfies the relation

$$f\left(\frac{5x-3y}{2}\right) = \frac{5f(x)-3f(y)}{2} \quad \forall x, y \in R$$

and $f(0) = 3$ and $f'(0) = 2$, then the period of $\sin f(x)$ is

- A) 2π B) π C) 3π D) 4π

6. If $f(x) = x^{\frac{1}{\ln x}}$ then set of all possible

values of x for which $f'(x) = 0$ is

- A) $(0, \infty)$ B) $(1, \infty)$ C) $(0, 1) \cup (1, \infty)$ D) $(0, 1)$

7. If $y = \sqrt{(a-x)(x-b)} - (a-b) \tan^{-1} \sqrt{\frac{a-x}{x-b}}$ ($a > b$)

then $\frac{dy}{dx} =$

- A) $\sqrt{\frac{a-x}{x-b}}$ B) $\sqrt{(a-x)(x-b)}$
C) 0 D) 1

8. If $x^2 + xy + 3y^2 = 1$ then $(x+6y)^3 \frac{d^2y}{dx^2}$ is

- A) 0 B) -12 C) 22 D) -22

9. If $y = \log\left(\frac{x}{a+bx}\right)^x$ then $x^3 \frac{d^2y}{dx^2} =$

A) $\left(x \frac{dy}{dx} - y\right)^2$ B) $\left(x \frac{dx}{dy} - x\right)^2$

C) $\left(x \frac{dy}{dx} + y\right)^2$ D) $\left(x \frac{dx}{dy} - y\right)^2$

10. Suppose f and g are functions having second derivatives

f'' and g'' every where, if

$f(x) \cdot g(x) = 1$ for all x and f', g' are

never zero then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)}$
equals

- A) $2 \frac{f'(x)}{f(x)}$ B) $\frac{2g'(x)}{g(x)}$
C) $-\frac{f'(x)}{f(x)}$ D) $\frac{g'(x)}{g(x)}$

11. If $t(1+x^2) = x$ and $x^2 + t^2 = y$ then at

$$x=2, \frac{dy}{dx} =$$

- A) $\frac{24}{5}$ B) $\frac{101}{125}$ C) $\frac{488}{125}$ D) $\frac{358}{125}$

12. If $x = \frac{1+t}{t^3}, y = \frac{3}{2t^2} + \frac{2}{t}$ satisfies

$$f(x) \left(\frac{dy}{dx} \right)^3 = 1 + \frac{dy}{dx} \text{ then } f(x) =$$

- A) x B) $\frac{x^2}{1+x^2}$ C) $x + \frac{1}{x}$ D) $x - \frac{1}{x}$

13. Let $f(x) = \frac{x^2 - x}{x^2 + 2x}$, then $\frac{d}{dx}(f^{-1}(x))$ is equal to

- A) $\frac{-3}{(1-x)^2}$ B) $\frac{3}{(1-x)^2}$
C) $\frac{-1}{(1-x)^2}$ D) $\frac{-7}{(1-x)^2}$

14. If $x + y = 3e^2$ then $\frac{d}{dx}(x^y) = 0$ for $x =$

- A) e B) e^2 C) e^e D) $2e^2$

MULTI ANSWER QUESTIONS

15. $f(x) = |x^2 - 3|x| + 2|$ then which of the following is/are true

- A) $f'(x) = 2x - 3$ for $x \in (0, 1) \cup (2, \infty)$
B) $f'(x) = 2x + 3$ for $x \in (-\infty, -2) \cup (-1, 0)$
C) $f'(x) = -2x - 3$ for $x \in (-2, -1)$
D) $f'(x) = 0$ for $x \in R$

16. If $y = x^{(\log x)^{\log(\log x)}}$, then $\frac{dy}{dx}$ is

- A) $\frac{y}{x} (\ln x^{\log x - 1}) + 2 \ln x \ln(\ln x)$
B) $\frac{y}{x} (\log x)^{\log(\log x)} (2 \log(\log x) + 1)$
C) $\frac{y}{x \ln x} [(\ln x)^2 + 2 \ln(\ln x)]$
D) $\frac{y \log y}{x \log x} [2 \log(\log x) + 1]$

17. If $f(x-y), f(x), f(y)$ & $f(x+y)$ are in A.P. $\forall x, y \in R$ and $f(0) \neq 0$ then

- A) $f(4) = f(-4)$ B) $f(2) + f(-2) = 0$
C) $f'(4) + f'(-4) = 0$ D) $f'(2) = f'(-2)$

18. If $f_n(x) = e^{f_{n-1}(x)}$ for all

$n \in N$ & $f_0(x) = x$, then $\frac{d}{dx}\{f_n(x)\}$ is

- a) $f_n(x) \frac{d}{dx}\{f_{n-1}(x)\}$ b) $f_n(x)f_{n-1}(x)$
c) $f_n(x)f_{n-1}(x) \dots f_2(x) \cdot f_1(x)$
d) zero

19. If $\frac{\cos^4 \theta}{x} + \frac{\sin^4 \theta}{y} = \frac{1}{x+y}$ (θ is constant)

$$\text{then } \frac{dy}{dx} =$$

- A) xy B) $\frac{y}{x}$
C) $\tan^2 \theta$ D) $(x^2 + y^2) \sec^2 \theta$

STATEMENT TYPE QUESTIONS

20. Statement 1 : Let $f(x) = x[x]$ & $[.]$ denotes greatest integral function,

DIFFERENTIATION

when x is not an integer, then $f'(x)$ is given by $[x]$

Statement 2 : $f'(x)$ does not exist for any $x \in \text{integer}$.

a) If both the statements are TRUE and STATEMENT 2 is the correct explanation of STATEMENT 1

b) If both the statements are TRUE but STATEMENT 2 is not the correct explanation of STATEMENT 1

c) If STATEMENT 1 is true and STATEMENT 2 is false

d) If STATEMENT 1 is false and STATEMENT 2 is true

COMPREHENSION TYPE QUESTIONS

Passage - I

$f(x)$ is a polynomial function $f: R \rightarrow R$

such that $f(2x) = f'(x) f''(x)$

21. The value of $f(3)$ is

- A) 4 B) 12 C) 15 D) 19

22. $f(x)$ is

- A) one - one and onto
B) one - one and into
C) many - one and onto
D) many one and into

23. Equation $f(x) = x$ has

- a) Three real and positive roots
b) Three real and negative roots
c) One real roots
d) Three real roots such that sum of roots is zero

Passage - II

$f: R \rightarrow R$ defined by ,

$f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$ for

all $x \in R$

24. The value of $f(1)$ is

- A) 2 B) 3 C) -1 D) 4

25. $f(x)$ is

- A) one - one and onto
B) one - one and into
C) many - one and onto
D) many one and into

26. The value of $f'(1) + x f''(2) + f'''(3)$ is

- A) 0 B) 1 C) 2 D) 3

JEE ADVANCED - VOL - III

MATRIX-MATCHING QUESTIONS

27. **COLUMN - I**

A) $y = f(x)$ be given by

$$x = t^5 - 5t^3 - 20t + 7$$

and $y = 4t^3 - 3t^2 - 18t + 3$. Then $\left(\frac{dy}{dx}\right)_{(t=1)}$ is

B) $P(x)$ be a polynomial of degree 4,

with $P(2) = -1$, $P'(2) = 0$, $P''(2) = 2$.

$$P'''(2) = -12 \text{ and } P^{iv}(2) = 24$$

then $P''(3)$

C) $y = \frac{1}{x}$, then $\frac{\frac{dy}{dx}}{\sqrt{1+y^4}}$ is

$$D) f\left(\frac{2x+3y}{5}\right) = \frac{2f(x)+3f(y)}{5}$$

and $f'(0) = p$; $f(0) = q$, then

$$f''(0) =$$

COLUMN - II

- P) 0 Q) $\frac{2}{5}$ R) 2 S) -1

INTEGER TYPE QUESTIONS

28. If $y = \sin^{-1} \left[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2} \right]$ and

$0 < x < 1$ then $\frac{dy}{dx}$ at $x = \frac{\sqrt{3}}{2}$ is

29. If $|x| < 1$ and $\frac{1-2x}{1-x+x^2} + \frac{2x-4x^3}{1-x^2+x^4} + \frac{4x^3-8x^7}{1-x^4+x^8} + \dots = \frac{1+ax}{1+bx+cx^2}$ then $a+b+c$ is

30. If $y = \frac{(ax+b)}{(x^2+c)}$, and

$(2xy' + y)y''' = k(xy'' + y')y''$, where a, b, c are constants and y' denote differentiation

DIFFERENTIATION

w.r.t x then the value of k is

31. If $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \log_e \sqrt{x+\sqrt{x^2+1}}$

and $x\left(\frac{dy}{dx}\right) + \log_e\left(\frac{dy}{dx}\right) = ky$ then k is

32. If $|a_1 \sin x + a_2 \sin 2x + \dots + a_n \sin nx| \leq |\sin x|$ for $x \in R$ then maximum value of

$|a_1 + 2a_2 + 3a_3 + \dots + na_n|$ is

33. If $y = \cos^{-1} \sqrt{\frac{\cos 3x}{\cos^3 x}}$ and $\frac{dy}{dx} = \sqrt{\frac{k}{\cos 2x + \cos 4x}}$, ($\sin x > 0$) then K is

SUBJECTIVE TYPE QUESTIONS

34. If $y = \left(\frac{1}{2}\right)^{n-1} \cos(n \cos^{-1} x)$, then prove that y satisfies the differential equation

$$(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + n^2 y = 0$$

35. If $x \in \left(0, \frac{\pi}{2}\right)$ then show that

$$\frac{d}{dx} \left(\cos^{-1} \left\{ \frac{7}{2}(1+\cos 2x) + \sqrt{(\sin^2 x - 48 \cos^2 x) \sin x} \right\} \right) = 1 - \frac{7 \sin x}{\sqrt{\sin^2 x - 48 \cos^2 x}}$$

36. If $\phi(x)$ be a polynomial function of the second degree. If $\phi(1) = \phi(-1)$ and a_1, a_2, a_3 are in AP, then prove that $\phi'(a_1), \phi'(a_2), \phi'(a_3)$ are in A.P.

KEY - LEVEL - V

- | | | | |
|----------|----------|-------------|----------|
| 1) B | 2) B | 3) A | 4) A |
| 5) B | 6) C | 7) A | 8) D |
| 9) A | 10) A | 11) C | 12) A |
| 13) B | 14) B | 15) A, B, C | 16) B, D |
| 17) A, C | 18) A, C | 19) B, C | 20) A |
| 21) B | 22) A | 23) D | 24) D |
| 25) C | 26) D | | |

27) A-Q; B-R; C-S; D-P

28) 2 29) 2 30) 3 31) 2

32) 1 33) 6

LEVEL - V

HINTS

SINGLE ANSWER QUESTIONS

1. $g'(x) = f(e^x) \cdot e^{f(x)} \cdot f'(x) + e^{f(x)} \cdot f'(e^x) \cdot e^x$
 $\therefore g'(0) = 8$

2. $y = \frac{(a-x)^{3/2} + (x-b)^{3/2}}{\sqrt{a-x} + \sqrt{x-b}}$

$$= \frac{(\sqrt{a-x} + \sqrt{x-b})(a-x - \sqrt{a-x}\sqrt{x-b} + x-b)}{\sqrt{a-x} + \sqrt{x-b}}$$

$$= a-b - \sqrt{a-x}\sqrt{x-b}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{a-x}} \sqrt{x-b} - \frac{1}{2\sqrt{x-b}} \sqrt{a-x}$$

$$= \frac{2x-a-b}{2\sqrt{a-x}\sqrt{x-b}}$$

3. $(\sin x)(\cos y) = 1/2$

$$\Rightarrow (\cos x)(\cos y) - \sin y \sin x \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = (\cot x)(\cot y)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -\cos^2 x \cot y - \cos^2 y \cot x \cdot \frac{dy}{dx}$$

now $\left(\frac{dy}{dx}\right)_{(\pi/4, \pi/4)} = 1$

$$\Rightarrow \left(\frac{d^2 y}{dx^2}\right)_{(\pi/4, \pi/4)} = -(2)(1) - (2)(1)(1) = -4$$

4. Given that $g^{-1}(x) = f(x) \Rightarrow x = g(f(x))$
 or $g'(f(x))f'(x) = 1$

$$\Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$

$$\Rightarrow g''(f(x))f'(x) = \frac{-f''(x)}{[f'(x)]^2} \Rightarrow g''(f(x)) = \frac{-f''(x)}{[f'(x)]^3}$$

5. Given $f\left(\frac{5x-3y}{2}\right) = \frac{5f(x)-3f(y)}{2}$

$$\Rightarrow f\left(\frac{5x-3y}{5-3}\right) = \frac{5f(x)-3f(y)}{5-3}, \text{ which}$$

satisfies section formula for abscissa on LHS and for ordinate of RHS. Hence $f(x)$ must be the linear function (as only straight line satisfies such section formula)

hence, $f(x) = ax + b$

$$\text{But } f(0) = 3 \Rightarrow b = 3, f'(0) = 2 \Rightarrow a = 2$$

$$6. \quad \ln(f(x)) = \frac{1}{\ln x} \cdot \ln x \Rightarrow \ln(f(x)) = 1$$

$$\Rightarrow f(x) = e \Rightarrow f'(x) = 0$$

$$7. \quad \text{Put } x = a \cos^2 \theta + b \sin^2 \theta$$

$$\Rightarrow \frac{dx}{d\theta} = (b-a) \sin 2\theta$$

$$\therefore \frac{dy}{dx} = \tan \theta = \sqrt{\frac{a-x}{x-b}}$$

$$8. \quad \text{From implicit derivatives}$$

$$\frac{dy}{dx} = \frac{-(2x+y)}{x+6y}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{-22(x^2 + xy + 3y^2)}{(x+6y)^3}$$

$$9. \quad y = x(\log x - \log(a+bx))$$

$$\Rightarrow \frac{dy}{dx} = \frac{a}{a+bx} + \log x - \log(a+bx)$$

$$\text{and } \frac{d^2y}{dx^2} = \frac{a^2}{x(a+bx)^2}$$

$$10. \quad \text{Given } f(x)g(x) = 1 \Rightarrow \log f(x) + \log g(x) = 0$$

$$\Rightarrow \frac{f'(x)}{f(x)} + \frac{g'(x)}{g(x)} = 0$$

again differentiating w.r.t. x we get

$$\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)} = \frac{-2g'(x)}{g(x)} = 2 \frac{f'(x)}{f(x)}$$

$$11. \quad \text{Given } t = \frac{x}{1+x^2} \Rightarrow \frac{dt}{dx} = \frac{1-x^2}{(1+x^2)^2}$$

$$x^2 + t^2 = y \Rightarrow \frac{dy}{dt} = 2x \left(\frac{(1+x^2)^3 + 1 - x^2}{1-x^4} \right)$$

$$\therefore \frac{dy}{dx} = \frac{2x((1+x^2)^3 + 1 - x^2)}{(1+x^2)^3}$$

$$12. \quad x = \frac{1}{t^3} + \frac{1}{t^2} \Rightarrow \frac{dx}{dt} = \frac{-(3+2t)}{t^4}$$

$$\text{and } \frac{dy}{dt} = \frac{-(3+2t)}{t^3}$$

$$\Rightarrow \frac{dy}{dx} = t \Rightarrow f(x) = \frac{1+t}{t^3} = x$$

$$13. \quad f(x) \text{ is not defined at } x = 0, -2$$

$$f(x) = 1 - \frac{3}{x+2}$$

$$f(f^{-1}(x)) = x$$

$$x = 1 - \frac{3}{f^{-1}(x)+2}$$

$$1-x = \frac{3}{f^{-1}(x)+2}$$

$$f^{-1}(x)+2 = \frac{3}{1-x}$$

Diff. w.r.t. (x)

$$\frac{d}{dx}(f^{-1}(x)) = \frac{3}{(1-x)^2}$$

$$14. \quad y = 3e^2 - x$$

$$f(x) = x^y = x^{3e^2-x}$$

$$\log f(x) = (3e^2 - x) \log(x)$$

Diff. w.r.t. ' x '

$$\frac{1}{f(x)} \cdot f'(x) = \frac{3e^2 - x}{x} - \log x$$

$$\text{Now } f'(x) = 0$$

$$3e^2 - x = x \log x$$

$$\therefore x = e^2$$

MULTI ANSWER QUESTIONS

$$15. \quad f(x) = |x^2 - 3|x| + 2|$$

$$= \begin{cases} |x^2 - 3x + 2|, & x \geq 0 \\ |x^2 + 3x + 2|, & x < 0 \end{cases}$$

$$= \begin{cases} x^2 - 3x + 2, & x^2 - 3x + 2 \geq 0, & x \geq 0 \\ -x^2 + 3x - 2, & x^2 - 3x + 2 < 0, & x \geq 0 \\ x^2 + 3x + 2, & x^2 + 3x + 2 \geq 0, & x < 0 \\ -x^2 - 3x - 2, & x^2 + 3x + 2 < 0, & x < 0 \end{cases}$$

$$= \begin{cases} x^2 - 3x + 2, & x \in [0, 1] \cup [2, \infty) \\ -x^2 + 3x - 2, & x \in (1, 2) \\ x^2 + 3x + 2, & x \in (-\infty, -2] \cup [-1, 0) \\ -x^2 - 3x - 2, & x \in (-2, -1) \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} 2x - 3, & x \in (0, 1) \cup (2, \infty) \\ -2x + 3, & x \in (1, 2) \\ 2x + 3, & x \in (-\infty, -2) \cup (-1, 0) \\ -2x - 3, & x \in (-2, -1) \end{cases}$$

16. $y = x^{(\log x)^{\log(\log x)}}$

$$\Rightarrow \log y = (\log x)(\log x)^{\log(\log x)} \rightarrow (1)$$

Taking log of both sides, we get

$$\Rightarrow \log(\log y) = \log(\log x) + \log(\log x) \log(\log x)$$

Diff. w.r.t x, we get

$$\frac{1}{\log y} \cdot \frac{1}{y} \frac{dy}{dx} = \frac{1}{x \log x} + \frac{2 \log(\log x)}{\log x} \cdot \frac{1}{x}$$

$$= \frac{2 \log(\log x) + 1}{x \log x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} \cdot \frac{\log y}{\log x} (2 \log(\log x) + 1) \Rightarrow D$$

Substituting the value of log y from (1), we get

$$\frac{y}{x} (\log x)^{\log(\log x)} (2 \log(\log x) + 1) \Rightarrow B$$

17. $f(x-y), f(x)f(y)$ & $f(x+y)$ are in A.P

$$\Rightarrow f(x+y) + f(x-y) = 2f(x)f(y) \rightarrow (1)$$

$$\forall x, y \in \mathbb{R}$$

putting $x=0, y=0$ in (1)

$$\text{we get } f(0) + f(0) = 2f(0)f(0)$$

$$\Rightarrow f(0) = 1 \quad (\because f(0) \neq 0)$$

Putting $x=0, y=x$

$$\text{We get } f(x) + f(-x) = 2f(0)f(x)$$

$$\Rightarrow f(x) = f(-x) \rightarrow (2)$$

$$\Rightarrow f(4) = f(-4), f(3) = f(-3)$$

Diff. (2) w.r.t x, $f'(x) + f'(-x) = 0$

$$\Rightarrow f'(4) + f'(-4) = 0 \text{ \& } f'(2) = -f'(-2)$$

18. $\frac{d}{dx}\{f_n(x)\} = \frac{d}{dx}\{e^{f_{n-1}(x)}\} = e^{f_{n-1}(x)} \frac{d}{dx}\{f_{n-1}(x)\}$

$$= f_n(x) \frac{d}{dx}\{f_{n-1}(x)\}$$

$$= f_n(x) \cdot \frac{d}{dx}\{e^{f_{n-2}(x)}\} = f_n(x) e^{f_{n-2}(x)} \frac{d}{dx}\{f_{n-2}(x)\}$$

$$= f_n(x) f_{n-1}(x) \frac{d}{dx}\{f_{n-2}(x)\}$$

$$= f_n(x) f_{n-1}(x) \dots f_2(x) \frac{d}{dx}\{f_1(x)\}$$

$$= f_n(x) \cdot f_{n-1}(x) \dots f_2(x) \frac{d}{dx}\{e^{f_0(x)}\}$$

Use $e^{f_0(x)} = f_1(x)$ & $f_0(x) = x$

19. $(x+y) \left(\frac{\cos^4 \theta}{x} + \frac{\sin^4 \theta}{y} \right) = 1$

$$\Rightarrow \left(\sqrt{\frac{x}{y}} \sin^2 \theta \right)^2 + \left(\sqrt{\frac{y}{x}} \cos^2 \theta \right)^2 - 2 \sin^2 \theta \cos^2 \theta = 0$$

$$\Rightarrow \tan^2 \theta = \frac{y}{x} \Rightarrow \frac{dy}{dx} = \tan^2 \theta = \frac{y}{x}$$

STATEMENT TYPE QUESTIONS

20. $f(x) = x[x] = \begin{cases} -x, & -1 \leq x < 0 \\ 0, & 0 \leq x < 1 \\ x, & 1 \leq x < 2 \\ 2x, & 2 \leq x < 3 \\ \dots & \dots \end{cases}$

$$f'(x) = \begin{cases} -1, & -1 < x < 0 \\ 0, & 0 < x < 1 \\ 1, & 1 < x < 2 \\ 2, & 2 < x < 3 \\ \dots & \dots \end{cases}$$

COMPREHENSION QUESTIONS

Passage - I

for - 21, 22, 23

Suppose degree of $f(x) = n$, then degree of $f'(x) = n-1$ and degree of $f''(x) = n-2$ so $n = n-1 + n-2$ Hence, $n = 3$

so put

 $f(x) = ax^3 + bx^2 + cx + d$. (where $a \neq 0$)From $f(2x) = f'(x) \cdot f''(x)$ we have $8ax^3 + 4bx^2 + 2cx + d$ $= (3ax^2 + 2bx + c)(6ax + 2b)$ $= 18a^2x^3 + 18abx^2 + (6ac + 4b^2)x + 2bc$

Comparing coefficients of terms, we have

 $18a^2 = 8a \Rightarrow a = 4/9$ $18ab = 4b \Rightarrow b = 0$ $2c = 6ac + 4b^2 \Rightarrow c = 0$ $d = 2bc \Rightarrow d = 0$ $\Rightarrow f(x) = \frac{4x^3}{9}$ which is clearly one one andonto. $\Rightarrow f(3) = 12$ Also, $\frac{4x^3}{9} = x \Rightarrow x = 0, x = \pm 3/2$

Hence sum of roots of equation is zero

Passage - II

for - 24, 25, 26

 $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$ Let $f'(1) = a, f''(2) = b, f'''(3) = c$ - (i) $f(x) = x^3 + ax^2 + bx + c$ $f'(x) = 3x^2 + 2ax + b$ $f'(1) = 3 + 2a + b$ - (ii) $f''(x) = bx + 2a$ $f''(2) = 12 + 2a$ - (iii) $f'''(x) = 6$ $f'''(3) = 6$ - (iv) $c = 6$

From (i), (ii), (iii)

$$f(x) = x^3 - 5x^2 + 2x + 6$$

MATRIX MATCHING QUESTIONS

27. a) $\left(\frac{dy}{dx}\right)_{(t=1)} = \frac{2}{5}$

b) Let us take

$$P(x) = a(x-2)^4 + b(x-2)^3 + c(x-2)^2 + d(x-2) + e$$

by using given data find a, b, c, d, e

c) Here

$$\sqrt{(1+y^4)} = \sqrt{\left(1 + \frac{1}{x^4}\right)} = \frac{\sqrt{1+x^4}}{x^2} \left(\because y = \frac{1}{x}\right)$$

$$\Rightarrow \frac{\sqrt{1+y^4}}{\sqrt{1+x^4}} = \frac{1}{x^2} \dots\dots\dots(1)$$

$$\text{but } y = \frac{1}{x} \therefore \frac{dy}{dx} = -\frac{1}{x^2} \dots\dots\dots(2)$$

$$\text{From 1 and 2 } \frac{\sqrt{1+y^4}}{\sqrt{1+x^4}} = -\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{\sqrt{(1+y^4)}} + \frac{dx}{\sqrt{(1+x^4)}} = 0$$

d) Obviously, $f(x)$ is a linear function.Also from $f'(0) = p$ and $f(0) = q$,

$$f(x) = px + q \Rightarrow f''(0) = 0$$

INTEGER TYPE QUESTIONS

28. $y = \sin^{-1} \left[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2} \right],$

where $0 < x < 1$

$$= \sin^{-1} \left[x\sqrt{1-(\sqrt{x})^2} - \sqrt{x}\sqrt{1-x^2} \right]$$

$$= \sin^{-1} x - \sin^{-1} \sqrt{x}$$

$$[\text{Using } \sin^{-1} x - \sin^{-1} y = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2})]$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-(\sqrt{x})^2}} \frac{d}{dx}(\sqrt{x})$$

$$= \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x}} \times \frac{1}{2\sqrt{x}}.$$

29. Consider $(1-x+x^2)(1-x^2+x^4).....\infty$

$$= \frac{(1+x+x^2)(1-x+x^2)(1-x^2+x^4)....\infty}{(1+x+x^2)}$$

$$= \frac{1-x^{2^n}+x^{2^{n+1}}}{1+x+x^2} = \frac{1}{1+x+x^2} (|x| < 1, n \rightarrow \infty)$$

now take log on both sides and differentiate

30. Given that $y(x^2+c) = ax+b$

Diff. w.r.t x

$$y'(x^2+c) + 2xy = a \dots\dots\dots (1)$$

Diff. again w.r.t x

$$y''(x^2+c) + y'2x + 2xy' + 2y = 0$$

$$\Rightarrow y''(x^2+c) + 2(2xy' + y) = 0 \dots\dots (2)$$

Diff. again w.r.t x

$$y'''(x^2+c) + 2xy'' + 2(2xy'' + 3y') = 0$$

$$\Rightarrow y'''(x^2+c) + 2xy'' + 2(2xy'' + 3y') = 0 \dots\dots (3)$$

Multiplying (2) by y''' & (3) by y'' and then subtracting we get

$$2(2xy' + y)y''' - 6(xy'' + y')y'' = 0$$

$$\Rightarrow (2xy' + y)y''' = 3(xy'' + y')y''$$

$$31. y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\log_e(x + \sqrt{x^2+1})$$

$$\Rightarrow y' = x + \frac{1}{2} \left[\frac{x^2}{\sqrt{x^2+1}} + \sqrt{x^2+1} \right] + \frac{1}{2\sqrt{x^2+1}}$$

$$= x + \frac{1}{2} \left[\frac{x^2+1}{\sqrt{x^2+1}} \right] + \frac{\sqrt{x^2+1}}{2} = x + \sqrt{x^2+1}$$

$$\text{Also } 2y = x^2 + x\sqrt{x^2+1} + \log_e(x + \sqrt{x^2+1})$$

$$= x(x + \sqrt{x^2+1}) + \log_e(x + \sqrt{x^2+1})$$

$$= xy' + \log_e y', \text{ hence proved.}$$

$$32. \left| \lim_{x \rightarrow 0} \frac{a_1 \sin x + a_2 \sin 2x + \dots + a_n \sin nx - 0}{x - 0} \right|$$

$$\leq \left| \lim_{x \rightarrow 0} \frac{\sin x - 0}{x - 0} \right|$$

$$\Rightarrow \left| (a_1 \sin x + a_2 \sin 2x + \dots + a_n \sin nx) \Big|_{x=0} \right|$$

$$\leq \left| (\sin x) \Big|_{x=0} \right|$$

$$\Rightarrow |a_1 + 2a_2 + \dots + na_n| \leq 1$$

33.

$$\frac{dy}{dx} = \frac{-1}{\sqrt{1 - \frac{\cos 3x}{\cos^3 x}}} \times \frac{1}{2\sqrt{\frac{\cos 3x}{\cos^3 x}}} \frac{d}{dx} (4 - 3 \sec^2 x)$$

$$= \frac{-\cos^3 x}{\sqrt{3 \cos x \sin^2 x}} \frac{1}{2\sqrt{\cos 3x}} (-3 \times 2 \sec x \cdot \sec x \cdot \tan x)$$

$$= \sqrt{\frac{6}{\cos 4x + \cos 2x}} \therefore K = 6$$

SUBJECTIVE TYPE QUESTIONS

$$34. y = (1/2^{n-1}) \cos(n \cos^{-1} x)$$

$$\frac{dy}{dx} = \frac{1}{2^{n-1}} \sin(n \cos^{-1} x) \left[\frac{-n}{\sqrt{1-x^2}} \right]$$

$$\Rightarrow (1-x^2) \left(\frac{dy}{dx} \right)^2 = \frac{n^2}{2^{2n-2}} \sin^2(n \cos^{-1} x)$$

$$= \frac{n^2}{2^{2n-2}} [1 - \cos^2(n \cos^{-1} x)]$$

$$= n^2 \left[\frac{1}{2^{2n-2}} - y^2 \right]$$

Diff. both sides w.r.t x

$$(1-x^2) 2 \frac{dy}{dx} \frac{d^2 y}{dx^2} - 2x \left(\frac{dy}{dx} \right)^2 = -2n^2 y \frac{dy}{dx}$$

$$\Rightarrow (1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + n^2 y = 0$$

$$35. y = \cos^{-1} \left\{ \frac{7}{2} (1 + \cos 2x) + \sqrt{(\sin^2 x - 48 \cos^2 x) \sin x} \right\}$$

$$= \cos^{-1} \left\{ (7 \cos x)(\cos x) + \sqrt{1 - 49 \cos^2 x} \sqrt{1 - \cos^2 x} \right\}$$

$$= \cos^{-1}(\cos x) - \cos^{-1}(7 \cos x)$$

$$(\because \cos x < 7 \cos x)$$

DIFFERENTIATION

$$= x - \cos^{-1}(7 \cos x)$$

now differentiating w.r.t x we get

$$= 1 + \frac{7 \sin x}{\sqrt{1 - 49 \cos^2 x}} = 1 + \frac{7 \sin x}{\sqrt{\sin^2 x - 48 \cos^2 x}}$$

$$36. \quad \phi(x) = ax^2 + bx + c$$

$$\because \phi(1) = \phi(-1) \Rightarrow a + b + c = a - b + c \Rightarrow b = 0$$

$$\phi(x) = ax^2 + c \Rightarrow \phi'(x) = 2ax$$

$$\therefore \phi'(a_1) = 2aa_1, \phi'(a_2) = 2aa_2, \phi'(a_3) = 2aa_3$$

$$\because a_1, a_2, a_3 \text{ are in AP}$$

$$\therefore \phi'(a_1), \phi'(a_2), \phi'(a_3) \text{ are also in AP.}$$